

## Homework 4 (Due: 5/29<sup>th</sup>)

M11202169

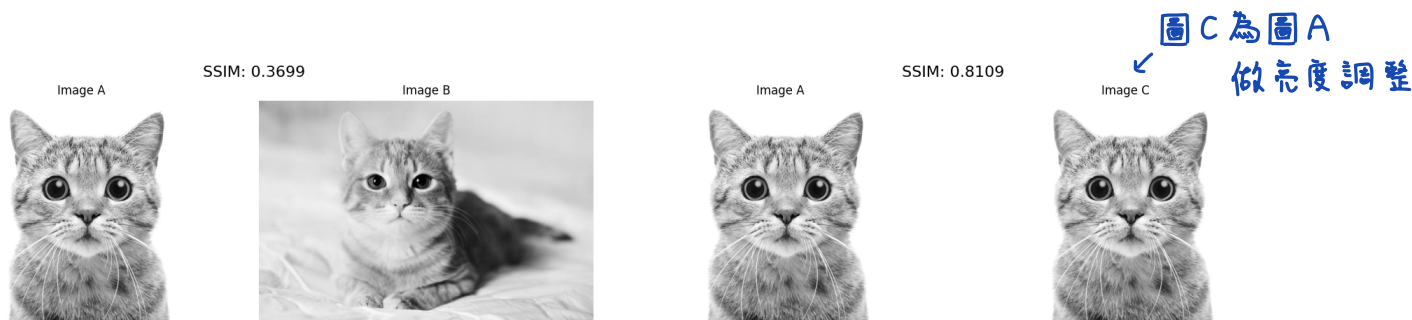
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(1) Write a Matlab or Python program to measure the structural similarity (SSIM) of two images A and B. The sizes of A and B are equivalent.

$$\text{SSIM}(A, B, c1, c2)$$

where  $c1$  and  $c2$  are some adjust constants.

The Matlab or Python code should be handed out by [NTUCool](#). (20 scores)



(2) Suppose that the probabilities of Chinese characters can be modeled as

$$P[n] = (\exp(0.002) - 1) \exp(-0.002n) \quad n = 1, 2, 3, \dots, 80000$$

(a) Determine the entropy of the Chinese characters. (b) Estimate the range of the coding length if we use the Huffman code to encode  $10^5$  Chinese characters using binary numbers. (c) Estimate the range of the coding length if we use the arithmetic code to encode  $10^5$  Chinese characters using binary numbers.

(15 scores)

$$\begin{aligned} \text{(a) entropy} &= - \sum_{n=1}^{80000} P[n] \ln P[n] \\ &= - \sum_{n=1}^{80000} (e^{0.002} - 1) e^{-0.002n} \ln[(e^{0.002} - 1) e^{-0.002n}] \\ &= - \sum_{n=1}^{80000} (e^{0.002} - 1) e^{-0.002n} [\ln(e^{0.002} - 1) - 0.002n] \\ &= - \ln(e^{0.002} - 1) \underbrace{\sum_{n=1}^{80000} (e^{0.002} - 1) e^{-0.002n}}_{1} + 0.002 (e^{0.002} - 1) \underbrace{\sum_{n=1}^{80000} n e^{-0.002n}}_{\frac{e^{-0.002}}{(1 - e^{-0.002})^2}} \\ &= - \ln(e^{0.002} - 1) + 0.002 (e^{0.002} - 1) \frac{e^{-0.002}}{(1 - e^{-0.002})^2} \\ &\approx 7.2146 \end{aligned}$$

$$\begin{aligned} \text{(b) } \text{ceil}\left(10^5 \times \frac{7.2146}{\ln 2}\right) &\leq \text{total coding length} \leq \text{floor}\left(10^5 \times \frac{7.2146}{\ln 2} + 10^5\right) \\ 1040847 &\leq \text{total coding length} \leq 1140846 \end{aligned}$$

$$\begin{aligned} \text{(c) } \text{ceil}\left(10^5 \times \frac{7.2146}{\ln 2}\right) &\leq \text{total coding length} \leq \text{floor}\left(10^5 \times \frac{7.2146}{\ln 2} + \log_2 2 + 1\right) \\ 1040847 &\leq \text{total coding length} \leq 1040848 \end{aligned}$$

(3) Suppose that  $x$  is a complex number. What are the constraints of  $\theta$  such that the multiplication of  $x$  and  $\exp(j\theta)$  required only 2 real multiplications?

(10 scores)

$$(a+jb)(c+jd) = e+jf \rightarrow \begin{bmatrix} e \\ f \end{bmatrix} = \begin{bmatrix} c & -d \\ d & c \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

specially, if  $c+jd = e^{j\theta} = \cos\theta + j\sin\theta$

$$\Rightarrow \begin{bmatrix} e \\ f \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

(i) if  $\sin\theta = \pm\cos\theta$ ,  $\theta = \frac{n\pi}{4}$ ,  $n=1,3,5,\dots$

$$\sin\theta = \cos\theta$$

$$\sin\theta = -\cos\theta$$

$$\begin{bmatrix} e \\ f \end{bmatrix} = \underbrace{\begin{bmatrix} \cos\theta & \cos\theta \\ \cos\theta & \cos\theta \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}}_{1\text{MUL}} + \underbrace{\begin{bmatrix} 0 & -2\cos\theta \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}}_{1\text{MUL}} \quad \begin{bmatrix} e \\ f \end{bmatrix} = \underbrace{\begin{bmatrix} \cos\theta & \cos\theta \\ \cos\theta & \cos\theta \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}}_{1\text{MUL}} + \underbrace{\begin{bmatrix} 0 & 0 \\ -2\cos\theta & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}}_{1\text{MUL}}$$

$\Rightarrow$  當  $\theta$  為  $\frac{n\pi}{4}$ ,  $n=1,3,5,\dots$ , 只需要 2MULs

(ii) if  $\sin\theta = \pm z^{-k}$  (trivial multiplication 情況),  $\theta = \sin^{-1}(\pm z^{-k})$ ,  $k=1,2,3,\dots$

$$\begin{bmatrix} e \\ f \end{bmatrix} = \begin{bmatrix} \cos\theta & \pm z^{-k} \\ \pm z^{-k} & \cos\theta \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}, \quad \begin{matrix} e = a\cos\theta \pm bz^{-k} \\ f = \pm az^{-k} + b\cos\theta \end{matrix} \quad \sim 2\text{MUL}$$

$\Rightarrow$  當  $\theta$  為  $\sin^{-1}(\pm z^{-k})$ ,  $k=1,2,3,\dots$ , 只需要 2MULs

(iii) if  $\cos\theta = \pm z^{-k}$  (trivial multiplication 情況),  $\theta = \cos^{-1}(\pm z^{-k})$ ,  $k=1,2,3,\dots$

$$\begin{bmatrix} e \\ f \end{bmatrix} = \begin{bmatrix} \pm z^{-k} & -\sin\theta \\ \sin\theta & \pm z^{-k} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}, \quad \begin{matrix} e = \pm az^{-k} - b\sin\theta \\ f = a\sin\theta \pm bz^{-k} \end{matrix} \quad \sim 2\text{MUL}$$

$\Rightarrow$  當  $\theta$  為  $\cos^{-1}(\pm z^{-k})$ ,  $k=1,2,3,\dots$ , 只需要 2MULs

(iv) if  $\sin\theta = \pm z^{\pm k} \cos\theta$ ,  $k=1,2,3,\dots$

$$\begin{bmatrix} e \\ f \end{bmatrix} = \begin{bmatrix} \cos\theta & \pm z^{\pm k} \cos\theta \\ \pm z^{\pm k} \cos\theta & \cos\theta \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}, \quad \begin{matrix} e = \cos\theta(a \pm bz^{\pm k}) \\ f = \cos\theta(\pm az^{\pm k} + b) \end{matrix} \quad \sim 2\text{MUL}$$

$\Rightarrow$  當  $\sin\theta = \pm z^{\pm k} \cos\theta$ ,  $k=1,2,3,\dots$ , 只需要 2MULs

(4) What is the complexity of the  $M \times N \times P$ -point 3D DFT? The deriving process should be given. (10 scores)

對  $M \times N$  二維矩陣的 DFT 運算複雜度為  $MN \log(MN)$ ,

總共有  $P$  個  $M \times N$  矩陣  $\Rightarrow$  運算複雜度為  $P \times (MN \log(MN))$ ,

接著視  $P$  為一維數組, 裡面元素為  $M \times N$  矩陣,

對  $P$  做 DFT  $\Rightarrow$  運算複雜度為  $MN \times (P \log P)$

總運算複雜度為:  $PMN \log(MN) + MN(P \log P) = MNP \log(MNP)$

The complexity of the  $M \times N \times P$ -point 3D DFT is  $\Theta(MNP \log(MNP))$

(5) How do we implement the 4-point DST-I with the least number of nontrivial multiplications? The number of real multiplications should also be shown.

$$X[m] = \sum_{n=1}^4 \sin\left(\frac{\pi}{5} mn\right) x[n] \quad \begin{matrix} m = 1, 2, 3, 4 \\ n = 1, 2, 3, 4 \end{matrix} \quad (15 \text{ scores})$$

$$\begin{bmatrix} X[1] \\ X[2] \\ X[3] \\ X[4] \end{bmatrix} = \begin{bmatrix} a & b & b & a \\ b & a & -a & -b \\ b & -a & -a & b \\ a & -b & b & -a \end{bmatrix} \begin{bmatrix} x[1] \\ x[2] \\ x[3] \\ x[4] \end{bmatrix} \quad a = 0.5878, \quad b = 0.9511$$

(Hint: we can convert it into two  $2 \times 2$  matrices.)

$$\begin{aligned} \begin{bmatrix} X[1] \\ X[3] \end{bmatrix} &= \begin{bmatrix} a & b \\ b & -a \end{bmatrix} \begin{bmatrix} x[1] + x[4] \\ x[2] + x[3] \end{bmatrix} \\ &= \underbrace{\begin{bmatrix} b & b \\ b & b \end{bmatrix} \begin{bmatrix} x[1] + x[4] \\ x[2] + x[3] \end{bmatrix}}_{1\text{MUL}} + \underbrace{\begin{bmatrix} a-b & 0 \\ 0 & -a-b \end{bmatrix} \begin{bmatrix} x[1] + x[4] \\ x[2] + x[3] \end{bmatrix}}_{2\text{MULs}} \Rightarrow 1+2 = 3\text{MULs} \end{aligned}$$

$$\begin{aligned} \begin{bmatrix} X[2] \\ X[4] \end{bmatrix} &= \begin{bmatrix} b & a \\ a & -b \end{bmatrix} \begin{bmatrix} x[1] - x[4] \\ x[2] - x[3] \end{bmatrix} \\ &= \underbrace{\begin{bmatrix} a & a \\ a & a \end{bmatrix} \begin{bmatrix} x[1] - x[4] \\ x[2] - x[3] \end{bmatrix}}_{1\text{MUL}} + \underbrace{\begin{bmatrix} -a+b & 0 \\ 0 & -a-b \end{bmatrix} \begin{bmatrix} x[1] - x[4] \\ x[2] - x[3] \end{bmatrix}}_{2\text{MULs}} \Rightarrow 1+2 = 3\text{MULs} \end{aligned}$$

總共需要  $3\text{MULs} + 3\text{MULs} = 6\text{MULs}$

- (6) Determining the numbers of real multiplications for the (a) 143-point DFT, (b) 195-point DFT, and the (c) 196-point DFT. (15 scores)

$$\begin{aligned} \text{(a)} \quad \text{MUL}_{143} &= 13 \text{MUL}_{11} + 11 \text{MUL}_{13} \\ &= 13 \times 40 + 11 \times 52 \\ &= 1092 * \end{aligned} \quad \begin{aligned} \text{(b)} \quad \text{MUL}_{195} &= 13 \text{MUL}_{15} + 15 \text{MUL}_{13} \\ &= 13 \times 40 + 15 \times 52 \\ &= 1300 * \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad \text{MUL}_{196} &= 49 \text{MUL}_4 + 4 \text{MUL}_{49} \\ &= 49 \times 0 + 4 \times (7 \text{MUL}_7 + 7 \text{MUL}_7 + 3 \times (7-1)(7-1)) \\ &= 4 \times 332 \\ &= 1328 * \end{aligned}$$

- (7) Derive the transform matrices of the (a) forward and (b) inverse 5-point NTTs where the prime number  $M$  is 11 and the value of  $\alpha$  should be as small as possible. (15 scores)

(a) 由於  $2^5 \% 11 = 10 \neq 1$  所以  $\alpha \neq 2$

$3^5 \% 11 = 1$  , 找到  $\alpha = 3$

$$\begin{bmatrix} F[0] \\ F[1] \\ F[2] \\ F[3] \\ F[4] \end{bmatrix} = \begin{bmatrix} 3^{0 \cdot 0} & 3^{1 \cdot 0} & 3^{2 \cdot 0} & 3^{3 \cdot 0} & 3^{4 \cdot 0} \\ 3^{0 \cdot 1} & 3^{1 \cdot 1} & 3^{2 \cdot 1} & 3^{3 \cdot 1} & 3^{4 \cdot 1} \\ 3^{0 \cdot 2} & 3^{1 \cdot 2} & 3^{2 \cdot 2} & 3^{3 \cdot 2} & 3^{4 \cdot 2} \\ 3^{0 \cdot 3} & 3^{1 \cdot 3} & 3^{2 \cdot 3} & 3^{3 \cdot 3} & 3^{4 \cdot 3} \\ 3^{0 \cdot 4} & 3^{1 \cdot 4} & 3^{2 \cdot 4} & 3^{3 \cdot 4} & 3^{4 \cdot 4} \end{bmatrix} \begin{bmatrix} f[0] \\ f[1] \\ f[2] \\ f[3] \\ f[4] \end{bmatrix} \pmod{M}$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 3 & 9 & 5 & 4 \\ 1 & 9 & 4 & 3 & 5 \\ 1 & 5 & 3 & 4 & 9 \\ 1 & 4 & 5 & 9 & 3 \end{bmatrix} *$$

有錯  
~~(b)~~  $9 \times 5 \% M = 1$   
 $\downarrow$   
 $\alpha^{-1}$

$$9 \times \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 3 & 9 & 5 & 4 \\ 1 & 9 & 4 & 3 & 5 \\ 1 & 5 & 3 & 4 & 9 \\ 1 & 4 & 5 & 9 & 3 \end{bmatrix} \pmod{M}$$

$$\Rightarrow \begin{bmatrix} 9 & 9 & 9 & 9 & 9 \\ 9 & 5 & 4 & 1 & 3 \\ 9 & 4 & 3 & 5 & 1 \\ 9 & 1 & 5 & 3 & 4 \\ 9 & 3 & 1 & 4 & 5 \end{bmatrix} *$$

(Extra): Answer the questions according to your student ID number.

(ended with (2, 7), (3, 8), (4, 9), (0, 5))

Q:  $\text{MUL}_{231} = ?$

$$\text{MUL}_{231} = 21 \text{MUL}_{11} + 11 \text{MUL}_{21}$$

$$= 21 \times 40 + 11 \times 62 \quad \text{查 page 371 表得知}$$

$$= 1522 *$$